

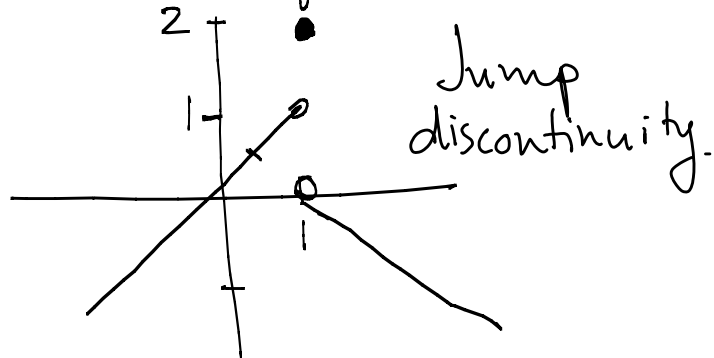
Unit 2.4 Continuity.

Note Title

2015/03/04

Look at the following functions:

$$\textcircled{1} f(x) = \begin{cases} x & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ 1-x & \text{if } x > 1 \end{cases}$$

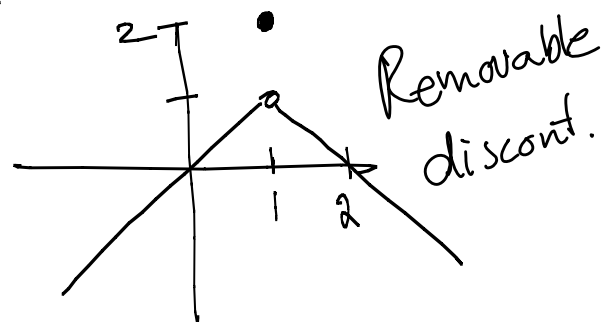


$$\lim_{x \rightarrow 1^-} f(x) = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = 0$$

$$f(1) = 2$$

$$\textcircled{2} f(x) = \begin{cases} x & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ 2-x & \text{if } x > 1 \end{cases}$$

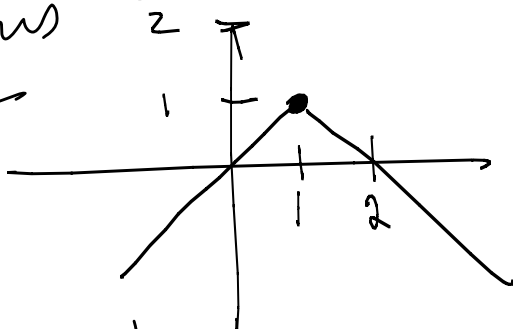


$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 1$$

$$f(1) = 2$$

$$\textcircled{3} f(x) = \begin{cases} x & \text{if } x < 1 \\ 1 & \text{if } x = 1 \\ 2-x & \text{if } x > 1 \end{cases}$$

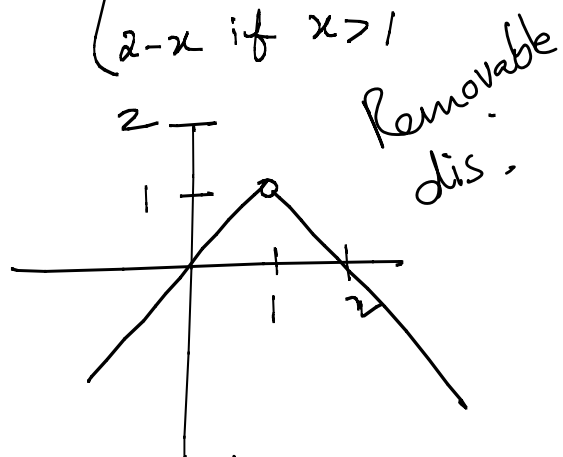
Continuous



$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = 1$$

$$f(1) = 1$$

$$\textcircled{4} f(x) = \begin{cases} x & \text{if } x < 1 \\ 2-x & \text{if } x > 1 \end{cases}$$



$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 1$$

$f(1)$ is not defined

f is continuous in a if: (Definition)

$$1. \lim_{x \rightarrow a} f(x) \text{ exists} \quad \left(\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) \right)$$

2. $f(a)$ is defined

$$3. \lim_{x \rightarrow a} f(x) = f(a).$$

Eg. Is f continuous in $x=1$?

$$f(x) = \begin{cases} 2x^2 & \text{if } x < 1 \\ -x+2 & \text{if } x \geq 1 \end{cases}$$

$$\left. \begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (2x^2) = 2 \\ \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (-x+2) = 1 \end{aligned} \right\} \begin{array}{l} \lim_{x \rightarrow 1} f(x) \\ \text{does not} \\ \text{exist} \end{array}$$

~~$f(1)$~~

f is discontinuous in $x=1$.

Eg. Find k such that f is continuous in $x=2$:

$$f(x) = \begin{cases} kx+4 & \text{if } x < 2 \\ 5 & \text{if } x = 2 \\ x^2+2k & \text{if } x > 2 \end{cases}$$

$$\left. \begin{aligned} \lim_{x \rightarrow 2^-} (kx+4) &= 2k+4 \\ \lim_{x \rightarrow 2^+} (x^2+2k) &= 4+2k \end{aligned} \right\} \Rightarrow 2k+4 = 4+2k.$$

For any value of k the $\lim_{x \rightarrow 2} f(x)$ exists

$$f(2) = 5 \Rightarrow 2k+4 = 5$$

$$\Rightarrow k = \frac{1}{2}$$

Continuity examples

$$1. \quad f(x) = \begin{cases} \frac{x^2-1}{x-1} & \text{if } x \neq 1 \\ 2 & \text{if } x = 1 \end{cases}$$

Is f continuous in $x=1$?

$$\left. \begin{aligned} \lim_{x \rightarrow 1} \frac{x^2-1}{x-1} &= \lim_{x \rightarrow 1} (x+1) = 2 \\ f(1) &= 2 \end{aligned} \right\} \text{The same}$$

$\Rightarrow f$ is cont in $x=1$.

$$2. \quad f(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0 \\ 3 & \text{if } x = 0 \end{cases}$$

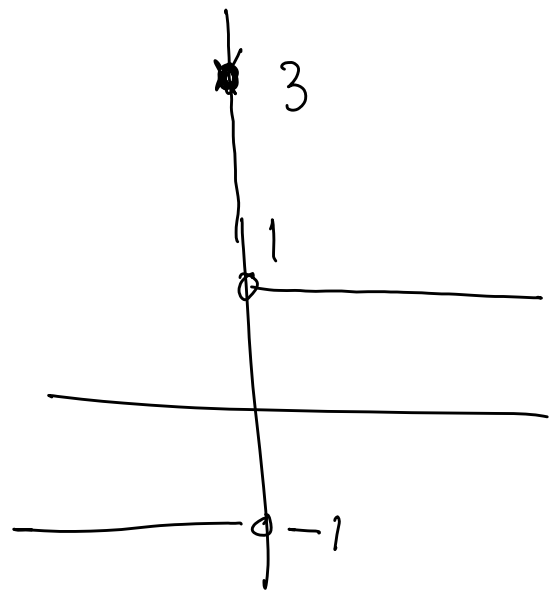
What happens in $x=0$?

$$\lim_{x \rightarrow 0^-} f(x) = -1$$

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

$\Rightarrow \lim_{x \rightarrow 0} f(x)$ does not exist,

$\Rightarrow f$ is discontinuous in $x=0$.



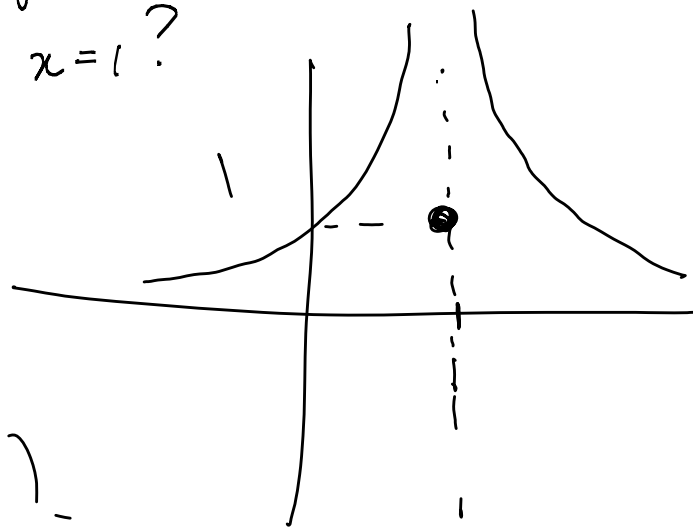
$$3. f(x) = \begin{cases} \frac{1}{(x-1)^2} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$$

What happens in $x=1$?

$$\lim_{x \rightarrow 1} \frac{1}{(x-1)^2} = \infty$$

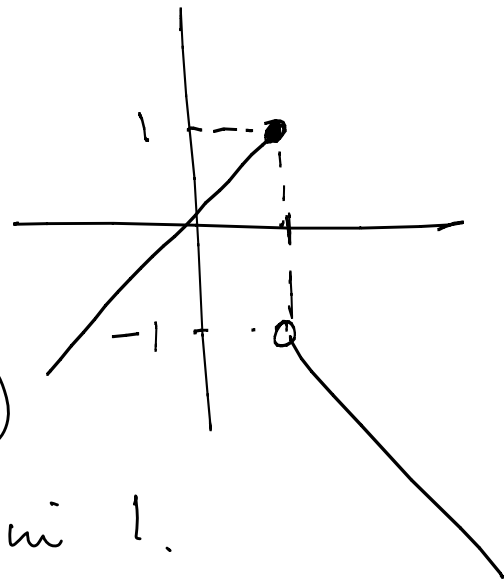
f is discontinuous in $x=1$

(Infinite discontinuity)



Left and right continuity.

$$1. \quad f(x) = \begin{cases} x & \text{if } x < 1 \\ 1 & \text{if } x = 1 \\ -x & \text{if } x > 1 \end{cases}$$

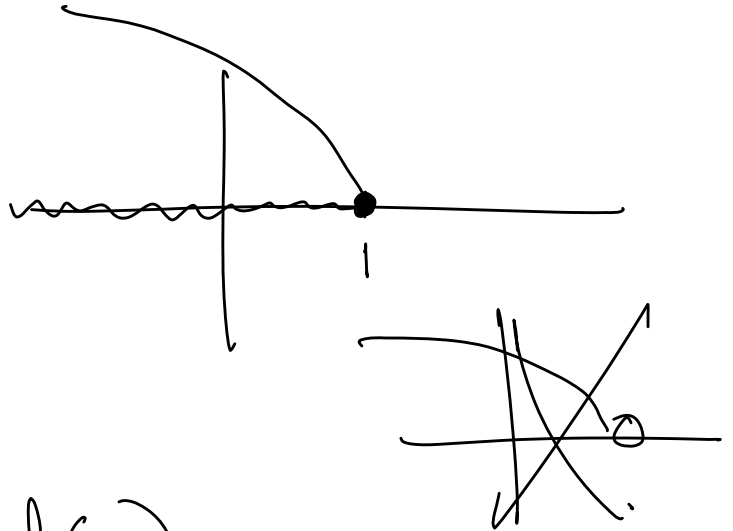


$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x = 1 = f(1)$$

$\Rightarrow f$ is left continuous in 1.

$$2. \quad f(x) = \sqrt{1-x}$$

$1-x \geq 0$
 $\Rightarrow x \leq 1$

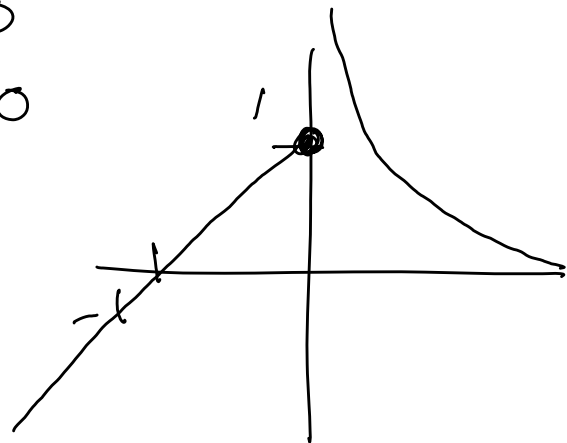


$$\lim_{x \rightarrow 1^-} f(x) = 0 = f(1)$$

$\Rightarrow f$ is left continuous in $x=1$.

$$3. \quad f(x) = \begin{cases} 1+x, & x < 0 \\ 1, & x = 0 \\ \frac{1}{x}, & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f(x) = 1 = f(0)$$



$$\lim_{x \rightarrow 3} \frac{\sin(x-3)}{\underbrace{x^2-9}_{(x-3)(x+3)}} = \lim_{x \rightarrow 3} \frac{\sin(x-3)}{x-3} \times \frac{1}{x+3}$$

$\swarrow$ $\searrow$ $\downarrow$
 $x-3 \rightarrow 0$ 1 6

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x} \quad \left(\frac{0}{\pi} \right)$$

$= 0$

$\lim_{x \rightarrow 0} \frac{\sin(\frac{x}{2})}{(\frac{x}{2})} = 1$
 3 positions the same